

Revision Questions

Question 1 [Total Marks: 15]

- a) Write a query to calculate the total amount (transaction table) spent by each customer across all their transactions.
- Use the **SUM()** window function to calculate the total amount spent **partitioned** by customer_id.
 - Join the required tables to match each transaction with the corresponding customer.
 - The output should include customer_id, first_name, last_name, and the total amount spent (as total_spent).

[4 Marks]

- b) Write a query to find the names of all products that have a price **higher than the average price** of all products.
- Use a **subquery** to calculate the average price of products from the products table.
 - The output should include product_name and price.

[3 Marks]

- c) Write a query to list all transactions along with the corresponding customer names and total amounts.
- Use **JOIN** to connect the transactions table with the customers table.
 - The output should include transaction_id, first_name, last_name, and total_amount.

[2 Marks]

d) Write a query to categorize each transaction based on its status.

- Use a **CASE** statement to create a new column called transaction_category that categorizes transactions as:
 - 'Completed' for transactions with status 'Completed'.
 - 'In Progress' for transactions with status 'Pending'.
 - 'Failed' for transactions with status 'Cancelled', else 'Unknown'.
- The output should include transaction_id, status, and transaction_category.

[3 Marks]

e) Write a query to calculate the running total of the total_amount for all transactions over time.

- Use the **SUM()** window function to calculate the running total.
- The running total should be **ordered by** transaction_date.
- The output should include transaction_id, transaction_date, total_amount, and the running total (as running_total).

[3 Marks]

Question 2 [Total Marks: 15]

- a) Write a query to calculate the **average** total amount spent by all customers, partitioned by the status of the transactions.
- Use the **AVG()** window function to calculate the average amount for each transaction status.
 - The output should include transaction_id, status, total_amount, and the average (as avg_total_amount).

[4 Marks]

- b) Write a query to find the names of all customers (customer table) who have not made any transactions.
- Use a **subquery** to identify customers whose customer_id is not present in the **transactions table**.
 - The output should include first_name and last_name.

[3 Marks]

- c) Write a query to list all products that have been sold in any transaction.
- Use a **JOIN** to connect the **products** and **transaction_items** tables.
 - The output should include product_name and quantity for each product sold.

[2 Marks]

d) Write a query to categorize products based on their stock_quantity.

- Use a **CASE** statement to categorize products as:
 - 'In Stock' if the stock_quantity is greater than 50.
 - 'Low Stock' if the stock_quantity is between 1 and 50.
 - 'Out of Stock' if the stock_quantity is 0, else 'Unknown'.
- The output should include product_name, stock_quantity, and stock_status.

[3 Marks]

e) Write a query to find the first and last transaction date for each customer.

- Use the **MIN()** and **MAX()** functions to find the first and last transaction date.
- The output should include customer_id, first transaction (as first_transaction_date), last transaction (as last_transaction_date).

[3 Marks]